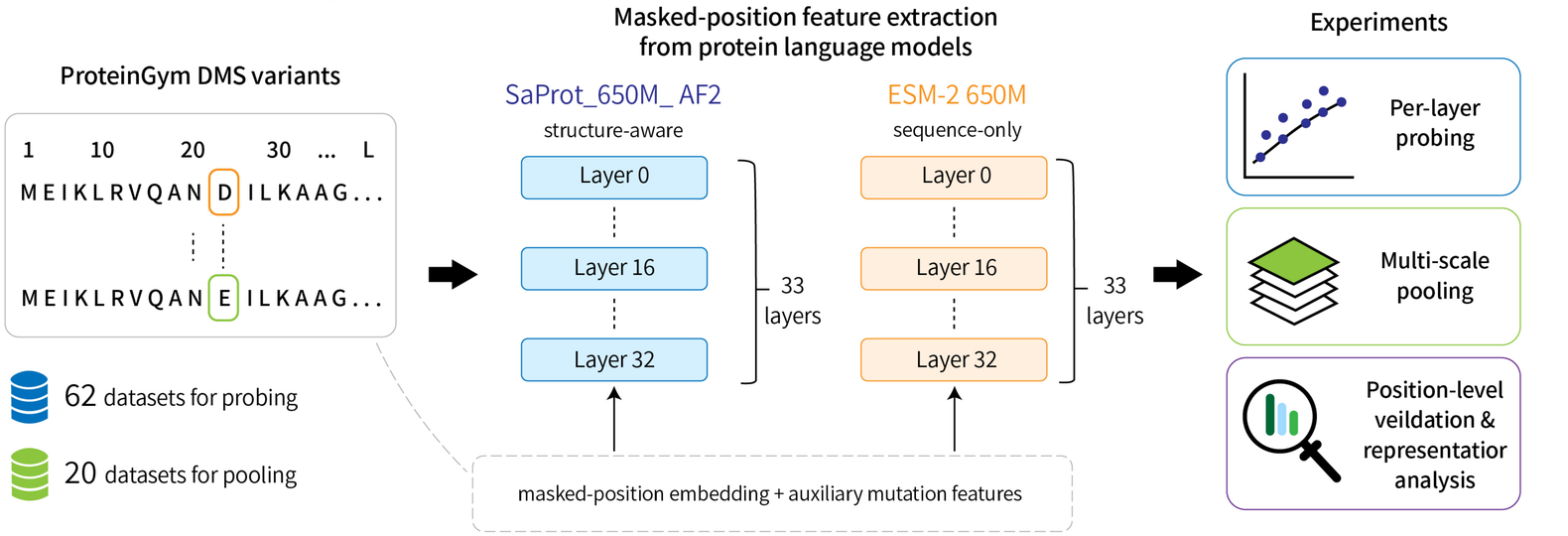
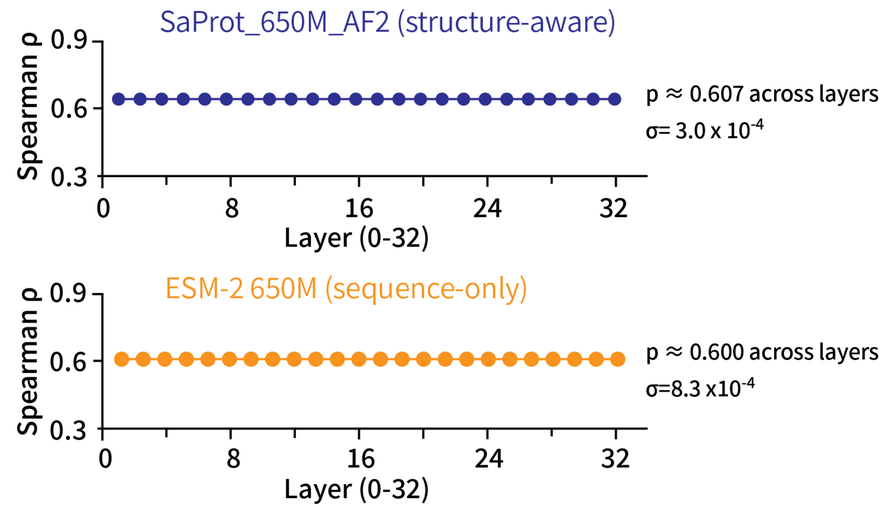
Layer invariance in protein language models for mutation effect prediction

Multi-scale embeddings do not improve prediction for structure-aware PLMs, and mainly linear readouts in standard ESM-2

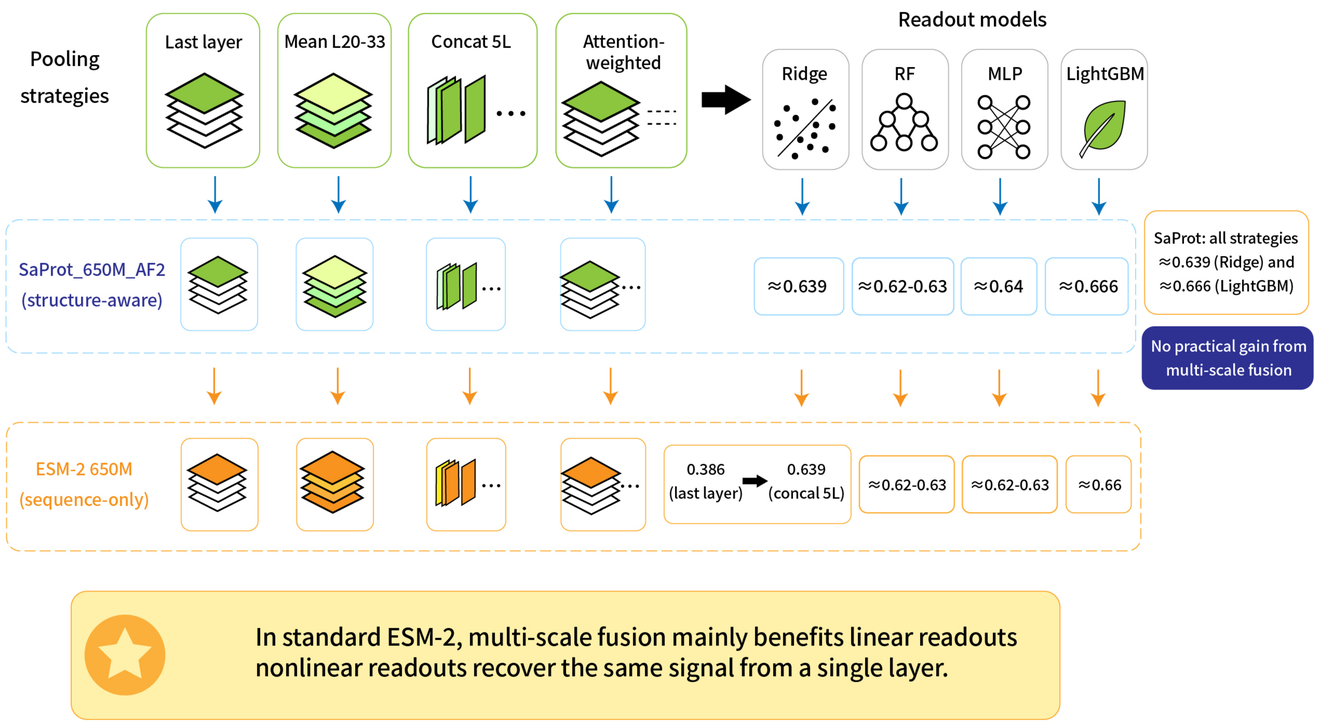
a Benchmark setup



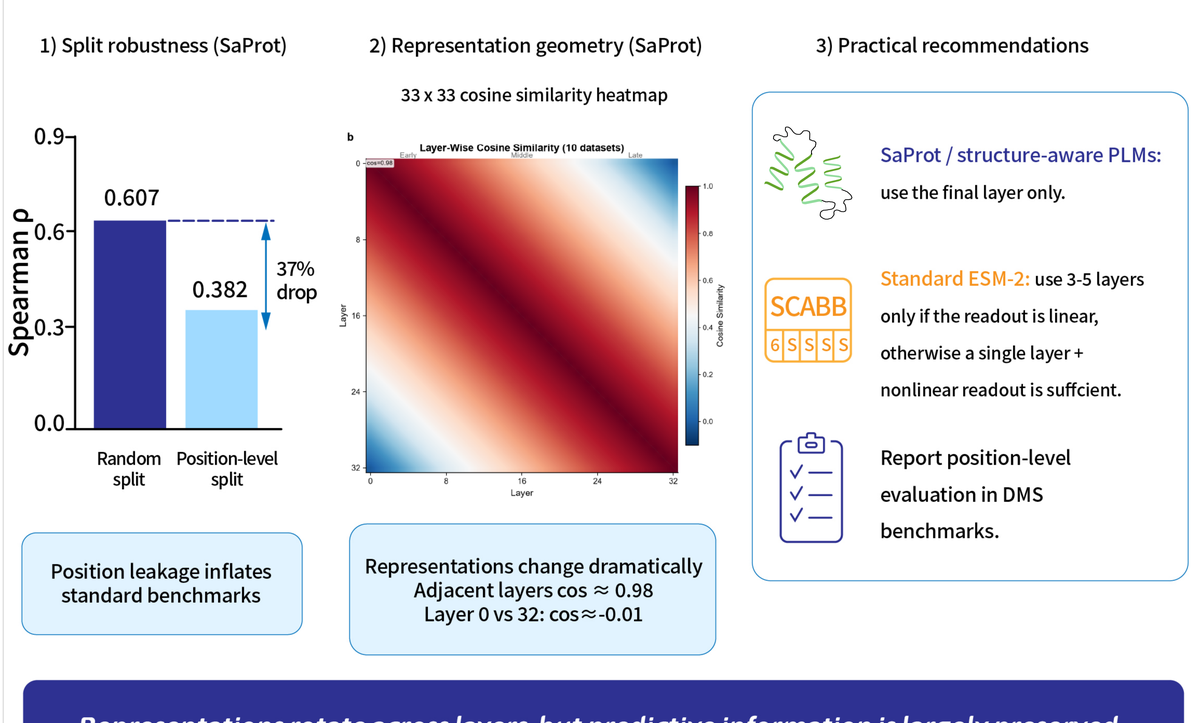
b Per-layer probing



c Multi-scale pooling reveals a model-dependent asymmetry



d Robustness and practical interpretation



Representations rotate across layers, but predictive information is largely preserved.